

ANIAK, AK, USA

WMO: **702320**Lat: **61.582N**Lon: **159.543W**Elev: **85**StdP: **14.65**Time Zone: **-9.00 (NAK)**Period: **94-19**WBAN: **26516**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.5	-28.2	-29.1	1.1	-18.8	-27.4	1.2	-16.4	24.3	34.8	20.5	31.0	0.4	120	0.713

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	73.5	58.2	70.5	57.3	66.4	55.7	60.7	70.3	58.8	67.5	57.1	65.0	6.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.6	68.4	63.3	54.6	63.5	61.4	53.6	61.3	60.5	26.8	71.0	25.5	67.5	24.4	65.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.2	13.6	-37.5	80.5	8.6	4.3	-43.7	83.6	-48.7	86.1	-53.5	88.5	-59.7	91.6		
(5)			-29.4	63.7	5.4	3.1	-33.3	65.9	-36.5	67.7	-39.5	69.5	-43.4	71.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.5	6.2	14.2	14.8	31.1	45.0	54.1	56.9	53.6	45.8	31.5	15.9	8.3
	DBStd	22.06	19.95	17.25	13.96	10.42	7.44	5.15	4.82	4.77	6.21	10.50	14.58	17.60
	HDD50	7279	1359	1001	1091	567	185	16	4	18	147	574	1024	1293
	HDD65	12223	1824	1421	1556	1017	621	328	256	355	576	1037	1474	1758
	CDD50	537	0	0	0	0	30	140	216	128	21	2	0	0
	CDD65	7	0	0	0	0	0	2	4	1	0	0	0	0
	CDH74	122	0	0	0	0	3	43	60	16	0	0	0	0
	CDH80	15	0	0	0	0	0	4	10	1	0	0	0	0
Wind	WSAvg	5.3	5.1	6.3	6.0	6.2	5.8	4.7	4.3	4.2	4.8	5.9	5.4	5.0
Precipitation	PrecAvg	23.8	1.6	1.5	1.3	1.1	1.4	1.9	2.6	3.3	3.0	2.2	2.0	1.8
	PrecMax	30.8	4.1	3.5	4.0	2.1	3.0	3.1	4.6	5.0	5.3	4.7	4.8	3.7
	PrecMin	18.1	0.2	0.2	0.1	0.2	0.4	0.9	1.1	1.5	0.6	0.3	0.2	0.6
	PrecStd	3.9	0.9	0.8	0.9	0.5	0.6	0.5	0.7	1.0	1.2	1.0	1.1	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.4	42.8	43.2	56.9	73.0	79.3	81.0	75.4	64.3	55.1	45.8	44.7
		MCWB	37.1	36.7	36.4	43.1	54.3	59.6	62.8	62.0	53.4	47.8	40.7	37.8
	2%	DB	39.4	39.2	39.2	52.3	66.3	73.4	73.5	69.3	61.1	51.8	42.6	39.5
		MCWB	34.7	34.6	33.9	41.2	51.1	57.0	59.5	59.1	51.8	45.7	38.5	35.1
	5%	DB	37.2	37.2	37.0	48.0	62.8	70.1	70.7	64.4	57.1	47.8	38.6	36.7
		MCWB	33.2	33.0	32.7	39.7	49.4	55.9	58.7	56.1	50.2	43.4	35.4	33.3
	10%	DB	35.5	36.1	35.2	45.2	57.4	65.8	66.4	62.7	54.8	45.3	35.9	33.8
		MCWB	32.3	32.5	31.6	37.9	46.3	54.3	57.1	55.4	49.4	41.3	33.5	31.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.3	37.4	37.5	44.1	55.8	62.2	65.6	63.6	55.2	49.2	41.5	39.6
		MCDB	43.3	41.1	42.3	54.4	69.9	74.9	78.0	73.0	61.8	54.8	45.2	43.9
	2%	WB	35.7	35.3	34.9	41.8	52.7	58.9	61.2	59.9	53.1	46.1	38.6	35.7
		MCDB	39.5	39.0	38.7	51.1	65.0	70.9	71.8	66.1	58.6	49.9	42.1	39.4
	5%	WB	34.0	34.1	33.3	39.9	50.0	56.8	59.3	57.8	51.4	43.8	35.6	33.6
		MCDB	37.7	37.3	37.1	47.6	60.9	67.5	68.5	63.5	55.9	47.5	38.4	37.0
	10%	WB	31.9	32.5	31.3	38.1	47.7	54.9	57.7	56.0	50.1	41.4	33.1	31.3
		MCDB	35.4	35.6	34.6	44.7	57.1	64.5	65.9	61.5	54.5	45.3	35.4	34.1
Mean Daily Temperature Range	5% DB	MDBR	12.8	14.6	17.0	15.8	16.9	17.4	14.5	12.9	13.0	10.7	11.8	13.3
		MCDBR	11.9	10.9	14.0	20.4	24.6	25.9	22.7	18.9	16.9	11.2	10.3	11.7
		MCWBR	9.6	8.9	10.6	12.3	13.1	12.7	11.4	11.0	10.5	7.6	8.4	9.6
	5% WB	MCDBR	11.8	10.9	12.8	18.1	22.3	22.3	20.2	15.9	14.7	10.5	9.9	11.3
		MCWBR	9.8	9.2	9.9	11.2	12.1	11.4	10.6	9.6	9.6	7.5	8.5	9.5
Clear-Sky Solar Irradiance	taub		0.222	0.270	0.279	0.344	0.370	0.365	0.363	0.370	0.320	0.290	0.230	N/A
	taud		2.057	2.208	2.256	2.167	2.311	2.382	2.429	2.411	2.531	2.419	2.071	N/A
	Ebn at Noon		156	224	268	266	269	273	271	258	255	219	147	N/A
	Edh at Noon		12	21	28	38	36	34	32	30	22	17	11	N/A
All-Sky Solar Radiation	RadAvg		105	324	917	1444	1613	1659	1390	1060	733	347	126	51
	RadStd		16	40	107	136	165	143	128	109	96	33	20	7